

BUPREN SUBLINGUAL TABLET

DESCRIPTION:

BUPREN 8MG/2MG: A white to off white hexagonal shaped tablet with a dimension of 11mmx10mm

BUPREN 2MG/0.5MG: A white to off white hexagonal shaped tablet with a dimension of 6.5mmx5.9mm

PHARMACODYNAMICS:

Buprenorphine is an opioid partial agonist/antagonist which binds to the μ (mu) and K (kappa) receptors of the brain. Its activity in opioid maintenance treatment is attributed to its slowly reversible properties with the μ receptors, which over a prolonged period, might minimize the need of addicted patients for drugs.

Naloxone is an antagonist at μ (mu)-opioid receptors. Because of its almost complete first pass metabolism, naloxone administered orally or sublingually has no detectable pharmacological activity. However, when administered intravenously to opioid dependent persons, the presence of naloxone in BUPREN produces marked opioid antagonist effect and opioid withdrawal, thereby deterring intravenous abuse.

PHARMACOKINETIC:

Buprenorphine

Absorption

Buprenorphine when taken orally undergoes first-pass metabolism with N-dealkylation and glucuroconjugation in the small intestine and the liver. The use of this medicinal product by the oral route is therefore inappropriate.

Peak plasma concentrations are achieved 90 minutes after sublingual administration. Plasma levels of buprenorphine increases with the sublingual dose.

Distribution:

The absorption of buprenorphine is followed by a rapid distribution phase.

Metabolism and elimination:

Buprenorphine is metabolized by 14 N-dealkylation and glucuroconjugation of the parent molecule and the dealkylated metabolite.

Elimination of buprenorphine is bi-or tri-exponential, with $t_{1/2}$ from plasma of 32 hours.

Buprenorphine is eliminated in the feces by biliary excretion of the glucuroconjugated metabolites (70%), the rest being eliminated in the urine.

Naloxone:

Absorption and distribution:

Plasma naloxone concentration are low and decline rapidly following a sublingual administration.

Metabolism and elimination: The drug is metabolized in the liver, primarily by glucuronide conjugation, and excreted in the urine. Naloxone has $t_{1/2}$ from plasma of 1.2 hours.

Hepatic impairment:

Lower initial BUPREN doses and cautious titration of dosage may be required in patients with severe hepatic dysfunction.

BUPREN is contraindicated in patients with severe hepatic dysfunction.

INDICATIONS:

Substitution treatment for opioid drug dependence, within a framework of medical, social and psychological treatment. The intention of the naloxone component is to deter intravenous misuse. Treatment is intended for use in adults and adolescents over 15 years of age who have agreed to be treated for addiction.

DOSAGE AND ADMINISTRATION:

Administration is sublingual.

Treatment must be under the supervision of a physical experienced in the management of opiate dependence/addiction.

BUPREN 2mg/0.5mg sublingual tablet containing 2 mg buprenorphine and 0.5 mg naloxone is referred to as the 2 mg tablets.

BUPREN 8mg/2mg sublingual tablet containing 8 mg buprenorphine and 2 mg naloxone is referred to as the "8 mg" tablets.

Physicians must warn patients that the sublingual route is the only effective and safe route of administration for this medicinal product, BUPREN sublingual tablets are to be placed under the tongue until dissolved, which usually requires 5 to 10 minutes. The dose is made up from BUPREN 2mg/0.5mg and BUPREN 8mg/2mg sublingual tablets, which may be taken all at the same time or in two divided portions; the second portion to be taken directly after the first portion has dissolved.

Adults:

Baseline liver function tests and documentation of viral hepatitis status is recommended prior to commencing therapy. Patient who are positive for viral hepatitis, on concomitant medicines and/ or have existing liver dysfunction are at risk of accelerated liver injury. Regular monitoring of liver function is recommended.

Induction:

Prior to treatment induction, consideration should be given to the type of opioid dependence (i.e. long-or-short-acting), the time since last opioid use and the degree of opioid dependence. To avoid precipitating withdrawal, induction with BUPREN or buprenorphine only tablets should be undertaken when objective and clear signs of withdrawal are evident.

Initiation therapy:

The recommended starting dose in one to two tablets of BUPREN 2mg/0.5mg sublingual tablets. An additional one to two tablets of the BUPREN 2mg/0.5mg sublingual tablets may be administered on day one depending on the individual patient's requirement.

- Opioid-dependent drug addicts who have not undergone withdrawal:

When treatment starts, the first dose of BUPREN should be taken when signs of withdrawal appear, but not less than 6 hours after the patients used opioids (eg, heroin; short acting opioids).

- Patients receiving methadone:

Before beginning BUPREN therapy, the dose of methadone must be reduced to a maximum of 30mg/day. The first dose of BUPREN should be taken when signs of withdrawal appear, but not less than 24 hours after the patient last used methadone. Buprenorphine may precipitate symptoms of withdrawal in patients dependent upon methadone.

Dosage adjustment and maintenance.

The dose of BUPREN should be increased progressively according to the clinical effect of the individual patient and should not exceed a maximum single daily dose of 24 mg. The dosage is

titrated according to reassessment of the clinical and psychological status of the patient and should be made in steps of 2-8mg.

During the initial of treatment, daily dispensing of buprenorphine is recommended. After stabilization, a reliable patient may be given a supply of BUPREN sufficient for several days of treatment. It is recommended that the amount of BUPREN be limited to 7 days according to local requirements.

Less than daily dosing:

After a satisfactory stabilization has been achieved the frequency of BUPREN dosing may be decreased to dosing every other day at twice the individual titrated daily dose. For example, a patient stabilized to receive a daily dose of 8 mg may be given 16 mg on alternate days, with no medication on the intervening days.

However, the dose given on any one day should not exceed 24mg. In some patients, after a satisfactory stabilization has been achieved, the frequency of BUPREN dosing may be decreased to 3 times a week (for example on Monday, Wednesday and Friday). The dose on Monday and Wednesday should be twice the individually titrated daily dose, and the dose on Friday should be three the individually titrated daily dose, with no dose on the intervening days. However, the dose given on any one day should not exceed 24mg. Patients requiring a titrated daily dose > 8mg/day may not find this regimen adequate.

Dosage reduction and termination of treatment:

After a satisfactory stabilization has been achieved, if the patient agrees, the dosage may be reduced gradually to a lower maintenance dose; in some favorable cases, treatment may be discontinued. The availability of the sublingual tablet in doses of 2mg/0.5mg and 8mg/2mg allows for a downward titration of dosage. Patients should be monitored following termination of treatment because of the potential for relapse.

Elderly:

No data is available on elderly patients.

Pediatrics:

BUPREN is not recommended for use in children below age 15 years due to lack of data on safety and efficacy.

Patients with impaired hepatic function:

The effect of hepatic impairment on the pharmacokinetics of buprenorphine and naloxone is unknown. Since both active medicines are extensively metabolized, the plasma levels will be expected to be higher in patients with moderate and severe hepatic impairment. It is not known whether both active substances are affected to the same extent.

As BUPREN pharmacokinetic may be altered in patients with hepatic insufficiency, therefore, lower initial doses and careful dose titration in patients with mild to moderate hepatic impairment are recommended.

Patients with impaired renal function:

Modification of the buprenorphine/naloxone dose is not generally required in patients with renal impairment. Caution is recommended when dosing patients with severe renal impairment (creatinine clearance < 30 ml/min)

ROUTE OF ADMINISTRATION:

Sublingual administration

CONTRAINDICATIONS:

BUPREN is contraindicated if the patient has:

- Hypersensitivity to buprenorphine, to naloxone, or to any of the excipients,
- Severe respiratory insufficiency
- Severe hepatic insufficiency,
- Acute alcoholism or *delirium tremens*.

WARNING AND PRECAUTIONS:

Due to the lack of data in adolescents (age 15-<18) BUPREN should be used only with caution in adolescents (age 15-18). Patients should be closely monitored during the switching period from buprenorphine or methadone to BUPREN since withdrawal symptoms have been reported.

Diversion:

Diversion refers to the introduction of buprenorphine into the illicit market either by patients or by individuals who obtain the medicinal product through theft from patients or pharmacies. This diversion may lead to new addicts using buprenorphine as the primary drug of abuse, with the risks of overdose, spread of blood borne viral infections, respiratory depression and hepatic injury. Because the naloxone in the combination tablet precipitated withdrawal in individuals dependent on heroin, methadone, or other agonist, BUPREN is expected to be less likely to be diverted for intravenous use.

Precipitated withdrawal:

When initiating treatment with buprenorphine, the physician must be aware of the partial agonist profile of buprenorphine and that it can precipitate withdrawal in opioid-dependent patients particularly if administered less than 6 hours after the last use of heroin or other short acting opioid, or if administered less than 6 hours after the last use of heroin or other short acting opioid, or if administered less than 24 hours after the last dose of methadone. Conversely withdrawal symptoms may also be associated with suboptimal dosing. The risk of serious undesirable effects such as overdosage or treatment dropout is greater if a patient is under dosage with BUPREN and continues to self medicate withdrawal symptoms with opioids, alcohol or other sedative-hypnotics in particular benzodiazepines.

Dependence:

Buprenorphine is a partial agonist at the mu-opiate receptor and chronic administration produces dependence of the opioid type.

Discontinuation of treatment may result in a withdrawal syndrome that may be delayed. BUPREN may cause drowsiness, particularly when taken together with alcohol or central nervous system depressants (such as tranquilizers, sedatives or hypnotics)

Respiratory depression:

When buprenorphine was used in combination with benzodiazepines or when buprenorphine was not used according to prescribing information can cause death due to respiratory depression. Death in association with concomitant administration of buprenorphine and other depressants such as alcohol or other opioids were reported.

Hepatitis, hepatic events:

The spectrum of abnormalities ranges from transient asymptomatic elevations in hepatic transaminases to cases reports of hepatic failure, hepatic necrosis, hepatorenal syndrome and hepatic encephalopathy. In many cases the presence of pre-existing liver enzyme abnormalities, infection with hepatitis B or hepatitis C virus, concomitant use of other potentially hepatotoxic medicines, and ongoing injecting drug use may have a causative or contributory role. When a hepatic event is suspected, further biological and etiological evaluation is required. Depending upon the findings, the medicinal product may be discontinued cautiously so as to prevent withdrawal symptoms and to prevent a return illicit drug use. If the treatment is continued, hepatic function should be monitored closely.

As buprenorphine is an opioid, pain as a symptom of a disease may be attenuated. Athletes must be aware that this medicine may cause a positive reaction to “antidoping” tests.

As with other opioids, caution is requested in patients using buprenorphine and having head injury, increased intracranial pressure, hypotension, prostatic hypertrophy or urethral stenosis. This product should be used with care in patients with: asthma or respiratory insufficiency, renal insufficiency, hepatic insufficiency. The concomitant use of monoamine oxidase inhibitors (MAOI) might produce exaggeration of the effect of opioid, based on experience with morphine. Medicines that inhibit the enzyme CYP3A4 may give rise to increased concentrations of buprenorphine. A reduction of the BUPREN dose may be needed. Patients already treated with CYP3A4 inhibit should have their dose of BUPREN titrated carefully since a reduced dose may be sufficient in these patients

Risks from Concomitant Use with Benzodiazepines

Profound sedation, respiratory depression, coma, and death may result from the concomitant use of Bupren with benzodiazepines. Observational studies have demonstrated that concomitant use of opioids and benzodiazepines increases the risk of drug-related mortality compared to use of opioids alone. Because of these risks, reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate. If the decision is made to newly prescribe a benzodiazepine and an opioid together, prescribe the lowest effective dosages and minimum durations of concomitant use. If the decision is made to prescribe a benzodiazepine in a patient already receiving an opioid, prescribe a lower initial dose of the benzodiazepine than indicated in the absence of an opioid, and titrate based on clinical response. If the decision is made to prescribe an opioid in a patient already taking a benzodiazepine, prescribe a lower initial dose of the opioid, and titrate based on clinical response. Follow patients closely for signs and symptoms of respiratory depression and sedation. Advise both patients and caregivers about the risks of respiratory depression and sedation when Bupren is used with benzodiazepines. Advise patients not to drive or operate heavy machinery until the effects of concomitant use of the benzodiazepine have been determined. Screen patients for risk of substance use disorders, including opioid abuse and misuse, and warn them of the risk for overdose and death associated with the use of benzodiazepines (See Interactions with Other Medicaments).

Serotonin Syndrome with Concomitant Use of Serotonergic Drugs

Cases of serotonin syndrome, a potentially life-threatening condition, have been reported during concurrent use of Bupren with serotonergic drugs (See Interactions with Other Medicaments). This may occur within the recommended dosage range. Serotonin syndrome symptoms may include mental-status changes (e.g. agitation, hallucinations, coma),

autonomic instability (e.g. tachycardia, labile blood pressure, hyperthermia), neuromuscular aberrations (e.g. hyperreflexia, incoordination) and/or gastrointestinal symptoms (e.g. nausea, vomiting, diarrhoea) and can be fatal (See Interactions with Other Medicaments). The onset of symptoms generally occurs within several hours to a few days of concomitant use, but may occur later than that. Discontinue Bupren if serotonin syndrome is suspected.

Adrenal Insufficiency

Cases of adrenal insufficiency have been reported with opioid use, more often following greater than one month of use. Presentation of adrenal insufficiency may include non-specific symptoms and signs including nausea, vomiting, decreased appetite, fatigue, weakness, dizziness, and low blood pressure. If adrenal insufficiency is suspected, confirm the diagnosis with diagnostic testing as soon as possible. If adrenal insufficiency is diagnosed, treat with physiologic replacement dosing of corticosteroids. Wean the patient off of the opioid to allow adrenal function to recover and continue corticosteroid treatment until adrenal function recovers. Other opioids may be tried as some cases reported use of a different opioid without recurrence of adrenal insufficiency. The information available does not identify any particular opioids as being more likely to be associated with adrenal insufficiency.

Sexual Function/Reproduction

Long term use of opioids may be associated with decreased sex hormone levels and symptoms such as low libido, erectile dysfunction, or infertility (See Postmarketing Experience).

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES:

BUPREN may cause drowsiness, dizziness, or impaired thinking, particularly when taken together with alcohol or central nervous system depressants. Therefore caution is advised when performing the above mentioned activities

DRUG INTERACTIONS:

BUPREN should not be taken together with:

- Alcoholic drinks or medications containing alcohol.

BUPREN should be used cautiously when co-administered with:

- Benzodiazepines: This combination may result in death due to respiratory depression of central origin.
- Other central nervous system depressants, other opioid derivatives (e.g. methadone, analgesics and antitussives), certain antidepressants, sedative H₁ –receptor antagonists, barbiturates, anxiolytics other than benzodiazepines, neuroleptics, clonidine and related substances: these combinations increase central nervous system depression. The reduced level of alertness can make driving and using machines hazardous.
- CYP3A4 inhibitors: Patients receiving BUPREN should be closely monitored, and may required dose-reduction if combined with potent CYP3A4 inhibitors (e.g. protease inhibitors like ritonavir, nelfinavir or indinavir or azole antifungals such as ketoconazole or itraconazole).
- CYP3A4 inducers: It is recommended that patients receiving BUPREN should be closely monitored if inducers (e.g. phenobarbital, carbamazepine, phenytoin, rifampicin) are co-administered.

Benzodiazepines

Due to additive pharmacologic effect, the concomitant use of opioids with benzodiazepines increases the risk of respiratory depression, profound sedation, coma and death. The concomitant use of opioids and benzodiazepines increases the risk of respiratory depression because of actions at different receptor sites in the central nervous system that control respiration. Opioids interact primarily at μ -receptors, and benzodiazepines interact at GABA_A sites. When opioids and benzodiazepines are combined, the potential for benzodiazepines to significantly worsen opioid-related respiratory depression exists.

Reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate (see Warnings and Precautions).

Limit dosage and duration of concomitant use of benzodiazepines and opioids, and follow patients closely for respiratory depression and sedation.

Serotonergic Drugs

The concomitant use of opioids with other drugs that affect the serotonergic neurotransmitter system has resulted in serotonin syndrome. If concomitant use is warranted, carefully observe the patient, particularly during treatment initiation and dose adjustment. Discontinue Bupren if serotonin syndrome is suspected. Examples of serotonergic drugs are selective serotonin reuptake inhibitors (SSRIs), serotonin and norepinephrine reuptake inhibitors (SNRIs), tricyclic antidepressants (TCAs), triptans, 5-HT₃ receptor antagonists, drugs that affect the serotonin neurotransmitter system (e.g. mirtazapine, trazodone, tramadol), monoamine oxidase (MAO) inhibitors (those intended to treat psychiatric disorders and also others, such as linezolid and intravenous methylene blue) (See Warnings and Precautions).

PREGNANCY AND LACTATION:

Towards the end of pregnancy high doses of buprenorphine may induce respiratory depression in the neonate even after a short period of administration. Long-term administration of buprenorphine during the last three months of pregnancy may cause a withdrawal syndrome in the neonate. BUPREN should not be used during pregnancy.

In case pregnancy occurs while on BUPREN treatment, the mother and the unborn child should be closely monitored and switched to buprenorphine if further treatment is required.

Lactation:

It is unknown whether naloxone is excreted in human breast milk. Buprenorphine and its metabolites are excreted in human breast milk. In rats buprenorphine has been found to inhibit lactation. Therefore, breast-feeding should be discontinued during treatment with BUPREN.

SIDE EFFECTS

The most common-treatment related undesirable effects reported during clinical trials with BUPREN were those related to withdrawal symptoms (e.g. abdominal pain, diarrhea, muscle aches, anxiety, sweating)

Infections and infestations Infection
Vaginitis
Blood and lymphatic system disorders
Anemia, thrombocytopenia, leucopenia, lymphadenopathy , leukocytosis

Immune system disorder Allergic reaction
Metabolism and nutrition disorders Peripheral oedema, weight decreased, hyperglycemia, hyperlipidemia, hypoglycemia
Psychiatric disorders Anxiety, nervousness, depression, libido decreased, thinking abnormal, drug dependence, amnesia, hostility, speech disorder, depersonalization, abnormal dream, apathy, euphoria
Nervous system disorders Insomnia, somnolence, dizziness, paresthesia, hypertonia, convulsion, agitation, tremor, hyperkinesia
Eye disorder Amblyopia, lacrimation disorders, mitosis, conjunctivitis
Cardiac disorder Myocardial infarction, angina pectoris, palpitation, tachycardia, bradycardia
Vascular disorder Vasodilation, hypertension, migraine, hypotension, heat stroke
Respiratory, thoracic and mediastinal disorders Rhinitis, pharyngitis, cough increased, dyspnoea, asthma, yawn
Gastrointestinal disorders Constipation, nausea, vomiting, dyspepsia, diarrhea, anorexia, flatulence, ulcerative stomatitis, tongue discolouration
Hepatobiliary disorder Liver function abnormal
Skin and subcutaneous tissue disorders Sweating, rash, pruritus, urticaria, exfoliative dermatitis, acne, skin nodule, alopecia, dry skin
Musculoskeletal, connective tissue and bone disorders Arthralgia, myalgia, leg cramps, arthritis
Renal and urinary disorders Albuminuria, urine abnormality, haematuria, kidney calculus, increased creatinine, urinary tract infection, dysuria, urinary retention
Reproductive system and breast disorders Impotence, amenorrhoea, abnormal ejaculation, menorrhagia, metrorrhagia
General disorders Withdrawal syndrome, headache, asthenia, fever, flu syndrome, malaise, accidental injury, chills, chest pain, abdominal pain, back pain, pain
Injury, poisoning and procedural complications Hypothermia

Buprenorphine used alone for treatment of opioid dependency has been associated with the following symptoms constipation, headache, insomnia, asthenia, drowsiness, nausea and vomiting, fainting and dizziness, orthostatic hypotension, and sweating. Other undesirable effects in association with buprenorphine alone are:

- Respiratory depression
- Hepatic necrosis and hepatitis
- Hallucinations,
- Cases of bronchospasm, angioneurotic edema and anaphylactic shock.

In case of intravenous misuse, local reactions, sometimes septic and potentially serious acute hepatitis have been reported.

In patients presenting with marked drug dependence, initial administration of buprenorphine can produce a withdrawal effect similar to that associated with naloxone.

Spontaneous abortion has been reported with both buprenorphine and buprenorphine-naloxone. It is not possible to establish a causal relationship since cases typically involve other drug use or risk factors for spontaneous abortion.

A neonatal abstinence syndrome has been reported among newborns of women who have received buprenorphine during pregnancy. The syndrome may be milder and more protracted than that from short acting full μ -opioid agonist. The nature of the syndrome may vary depending upon the mother's drug use history.

Postmarketing Experience:

Serotonin syndrome (See Warnings and Precautions)

Adrenal insufficiency (See Warnings and Precautions)

Androgen deficiency: Cases of androgen deficiency have occurred with chronic use of opioids. Chronic use of opioids may influence the hypothalamic-pituitary-gonadal axis, leading to androgen deficiency that may manifest as low libido, impotence, erectile dysfunction, amenorrhea, or infertility. The causal role of opioids in the clinical syndrome of hypogonadism is unknown because the various medical, physical, lifestyle, and psychological stressors that may influence gonadal hormone levels have not been adequately controlled for in studies conducted to date. Patients presenting with symptoms of androgen deficiency should undergo laboratory evaluation.

Infertility: Chronic use of opioids may cause reduced fertility in females and males of reproductive potential. It is not known whether these effects on fertility are reversible.

OVERDOSAGE:

In the event of overdose, general supportive measure should be instituted, including close monitoring of respiratory and cardiac status of the patient. The major symptoms requiring intervention is respiratory depression, which could lead to respiratory arrest and death. If the patient vomits, care must be taken to prevent aspiration of the vomitus.

Treatment

Symptomatic treatment of respiratory depression, and standard intensive care measures, should be implemented. A patent airway and assisted or controlled ventilation must be assured. The patients should be transferred to an environment within which full resuscitation facilities are available. Use of an opioid antagonist (i.e. naloxone) is recommended, despite the modest effect it may have in

reversing the respiratory symptoms of buprenorphine compared with its effects on full agonist opioid agents.

The long duration of action of BUPREN should be taken into consideration when determining of the treatment and medical surveillance needed to reverse the effects of an overdose.

PRESENTATION: Sublingual tablet in blister packs of 7s. Box of 28 or 7 tablets.

STORAGE: Store below 30°C

MANUFACTURED BY:

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5 1/2 Mile off Jalan Meru,
41050 Klang, Selangor

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